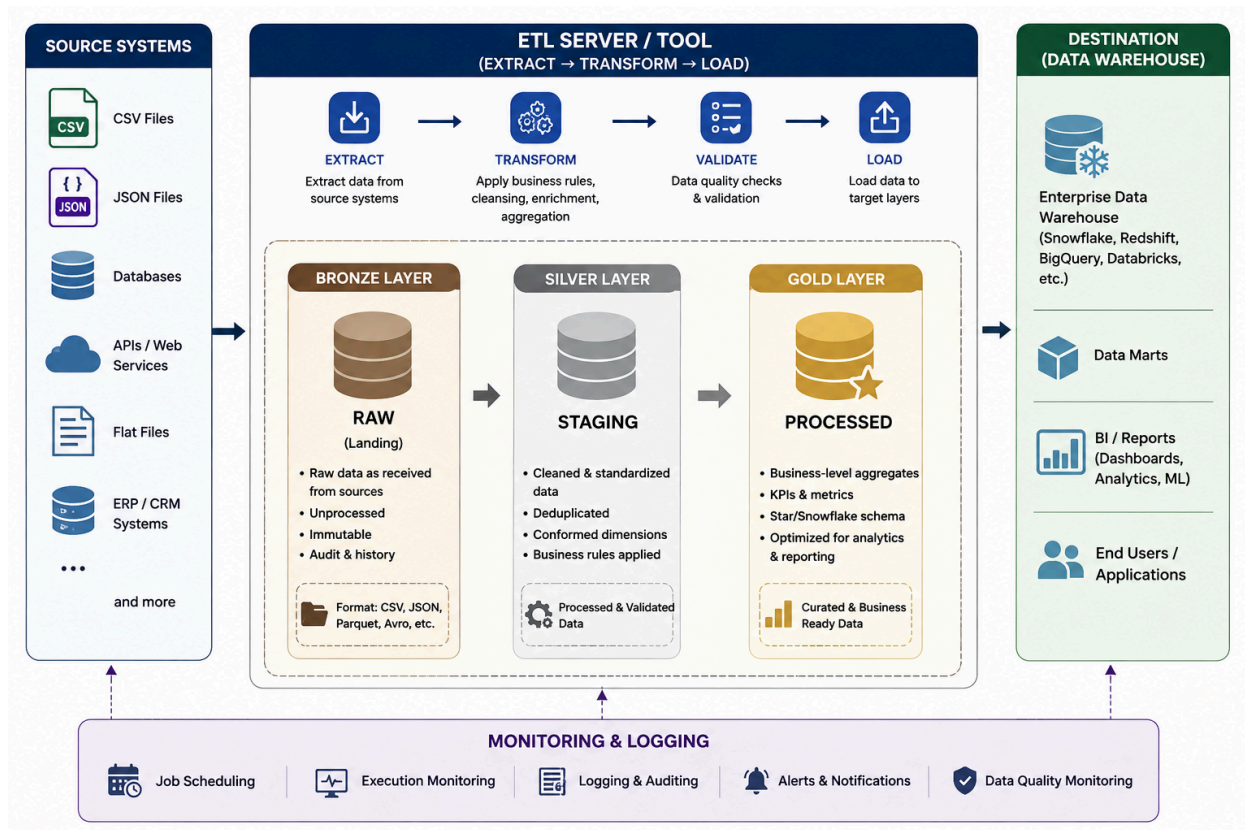




ETL / ELT Processes

1. ETL vs ELT

- What is ETL?
- What is ELT?
- ETL architecture
- ELT architecture
- ETL vs ELT comparison
- When to use ETL
- When to use ELT
- Batch vs real-time pipelines
- Full load vs incremental load
- CDC — Change Data Capture

2. Overview of ETL / Data Integration Tools

- Informatica
- Qlik Talend

- Fivetran
- Matillion
- Azure Data Factory
- AWS Glue
- Apache Airflow
- dbt
- Snowflake-native transformation approaches

3. Data Extraction from Source Systems

- Relational databases
- CSV / Excel files
- APIs
- JSON / XML
- Cloud storage
- Application databases
- Full extraction
- Incremental extraction
- Timestamp-based extraction
- Watermark-based extraction
- CDC-based extraction

4. Data Transformation Techniques

- Data type conversion
- Filtering
- Sorting
- Joining
- Aggregation
- Deduplication
- Lookup transformation
- Derived columns
- Conditional transformations
- String transformations
- Date/time transformations
- NULL handling
- Standardization
- Data enrichment
- Business-rule implementation

5. Data Loading Strategies

- Full load
- Incremental load

- Append load
- Upsert / Merge
- Truncate and load
- SCD-based loading
- Partition-based loading
- Parallel loading
- Batch loading
- Micro-batch loading
- Idempotent loading

6. Error Handling and Logging

- Data validation errors
 - Transformation errors
 - Connection failures
 - Duplicate records
 - Constraint violations
 - Rejected records
 - Error tables
 - Audit tables
 - ETL execution logs
 - Row-count validation
 - Source-to-target reconciliation
 - Retry mechanisms
 - Failure notifications
 - Restartability
-

Data Integration, Quality and BI Concepts

7. Data Integration

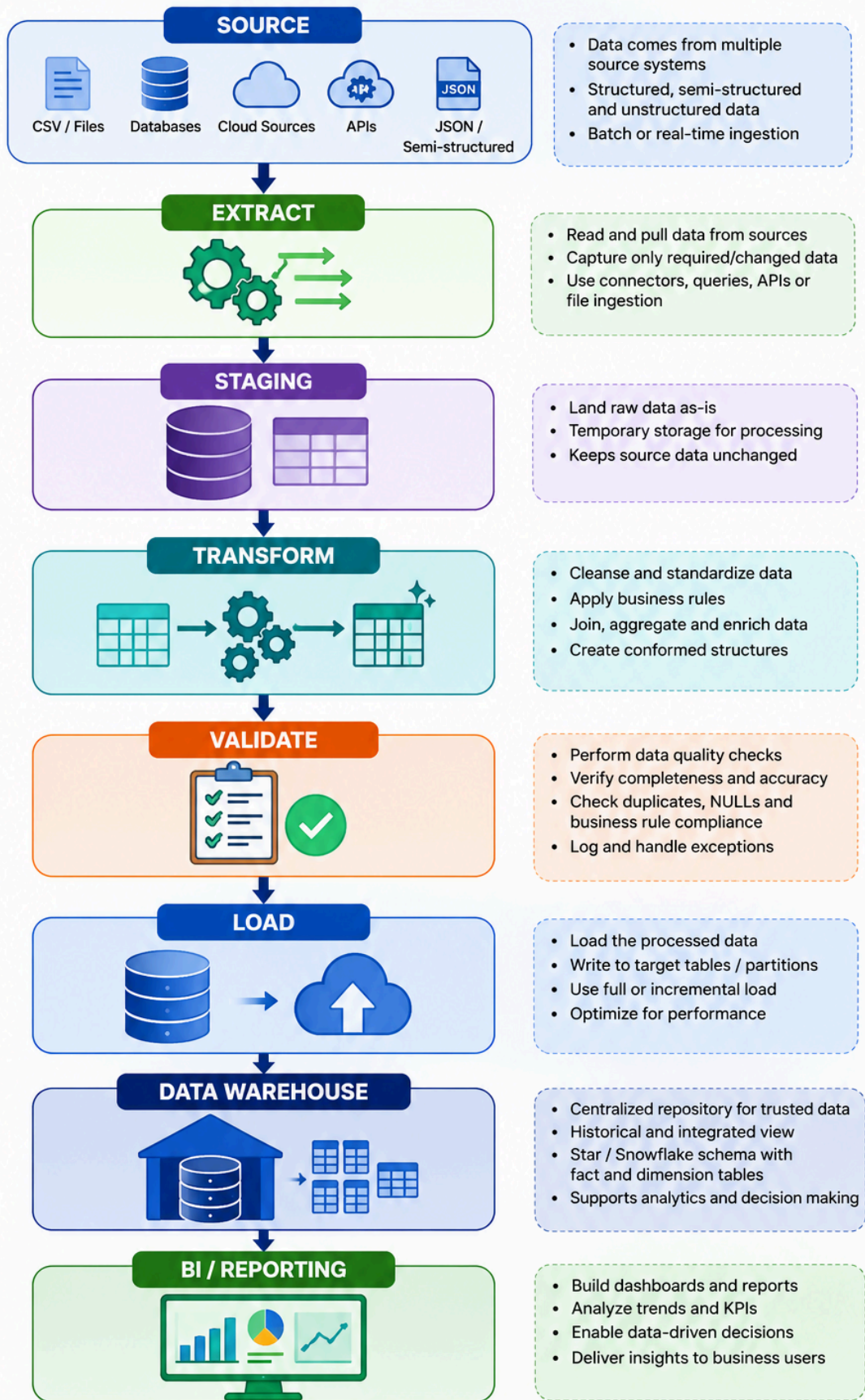
Core Concepts

- What is data integration?
- Source-to-target mapping
- Data pipelines
- Batch integration
- Real-time integration
- API integration
- Database integration

- File-based integration
- Cloud integration
- Data orchestration

Data Pipeline Flow – From Source to Insights

Build → Integrate → Transform → Deliver



DATA PIPELINE FLOW



8. Data Profiling and Cleansing

Data Profiling

- Column profiling
- Data type profiling
- NULL analysis
- Distinct-value analysis
- Frequency analysis
- Min / Max analysis
- Pattern analysis
- Outlier identification
- Data distribution

Data Cleansing

- Standardizing values
- Removing unwanted characters
- Correcting data types
- Handling invalid dates
- Standardizing names
- Standardizing addresses
- Handling NULL values
- Removing duplicates
- Correcting inconsistent categories

9. Handling Duplicates and Missing Data

Duplicates

- Identifying duplicates
- Exact duplicates
- Business-key duplicates
- Duplicate detection using `GROUP BY`
- Duplicate detection using `ROW_NUMBER()`
- Keeping the latest record
- Keeping the first record
- Removing duplicates safely

Missing Data

- NULL vs empty string
- Missing-value profiling
- Default values
- Imputation
- Forward filling
- Backward filling
- Business-rule-based replacement
- Rejecting incomplete records

Example:

```
SELECT
  CustomerID,
  CustomerName,
  COALESCE(City, 'Unknown') AS City
FROM Customer;
```

Required Labs

LAB 1 — Compute Management Lab

Objective

Understand how compute resources are used to execute transformation and analytical workloads.

Topics

- Compute resources
- CPU and memory
- Warehouses / clusters
- Scaling compute
- Small vs large workloads
- Query execution
- Cost vs performance
- Auto-suspend / auto-resume
- Workload isolation
- Query optimization

Suggested Snowflake Lab

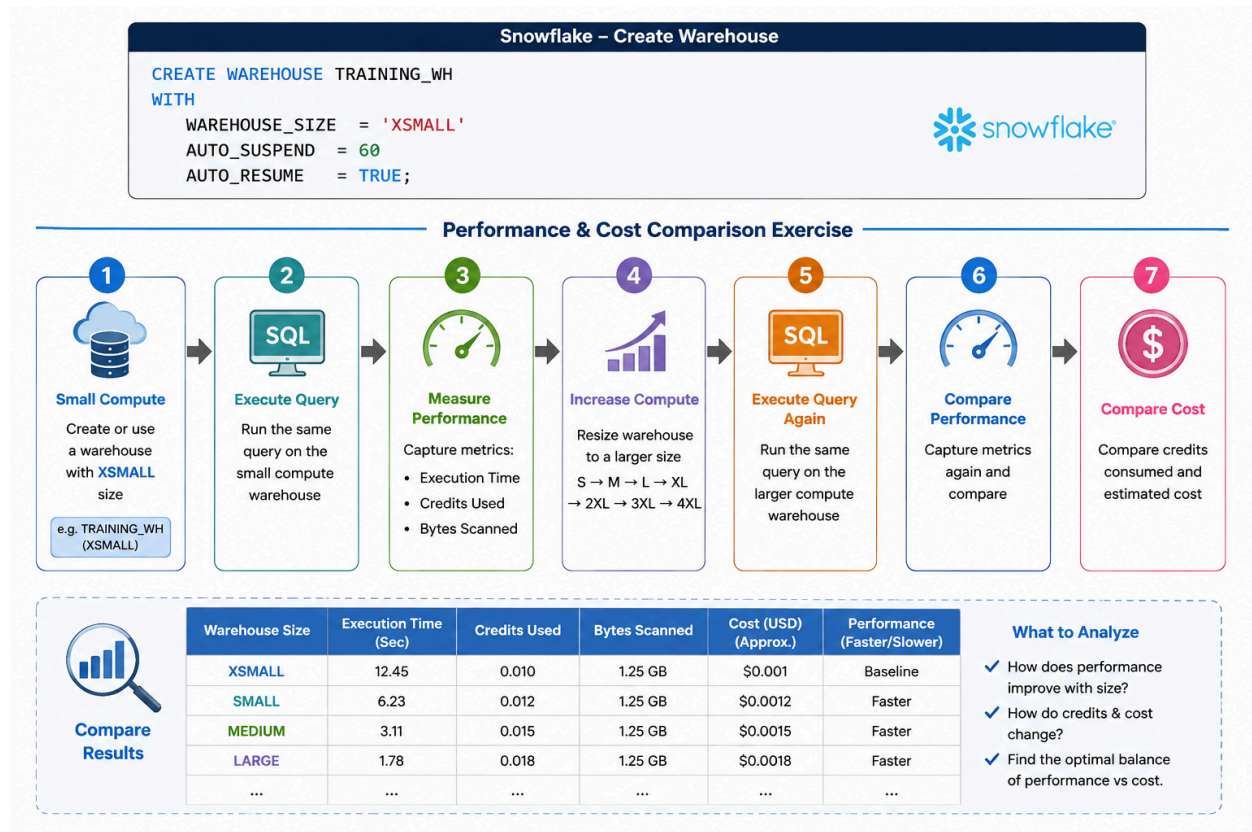
CREATE WAREHOUSE TRAINING_WH

WITH

WAREHOUSE_SIZE = 'XSMALL'

AUTO_SUSPEND = 60

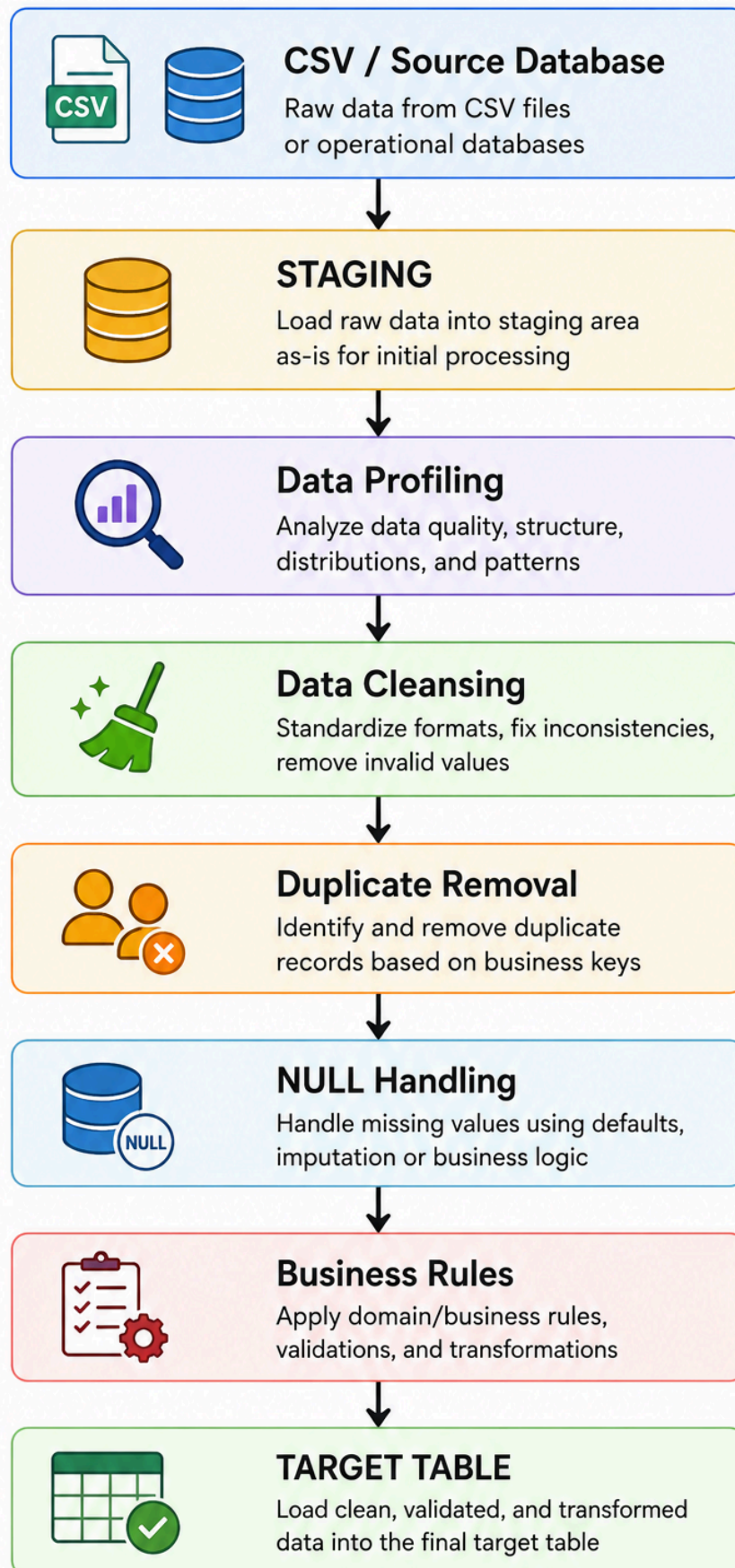
AUTO_RESUME = TRUE;



LAB 2 — Data Integration + Quality Lab

Objective

Build a small end-to-end pipeline that extracts, transforms, cleanses and loads data.



Tasks

- Load source data
- Profile the source
- Identify NULL values
- Identify duplicates
- Standardize values
- Remove duplicates
- Handle missing values
- Validate records
- Load clean data
- Create an audit summary

Expected Output

Source Records : 10,000
Valid Records : 9,650
Duplicate Records : 200
Rejected Records : 100
Missing-value Fixed : 50
Final Target Records : 9,650

LAB 3 — Query + Visualization Lab

Objective

Convert warehouse data into analytical queries and business dashboards.

Part A — SQL Querying

Students should practice:

- SELECT
- WHERE
- GROUP BY
- HAVING
- ORDER BY
- JOIN
- CASE
- Subqueries
- CTEs

- Window functions
- Aggregations
- Date functions
- Ranking
- Running totals
- Percentage calculations

Example Business Questions

1. What is total sales by region?
 2. Which products generate the highest revenue?
 3. What are the monthly sales trends?
 4. Who are the top 10 customers?
 5. Which region has the highest average order value?
 6. What percentage of total revenue comes from each category?
 7. Which products have declining sales?
 8. What is the month-over-month revenue growth?
 9. Which customers have not purchased recently?
 10. What are the top-performing sales representatives?
-

Part B — Visualization

Connect the query results to a BI/visualization tool and create:

KPI Cards

- Total Revenue
- Total Orders
- Total Customers
- Average Order Value

Charts

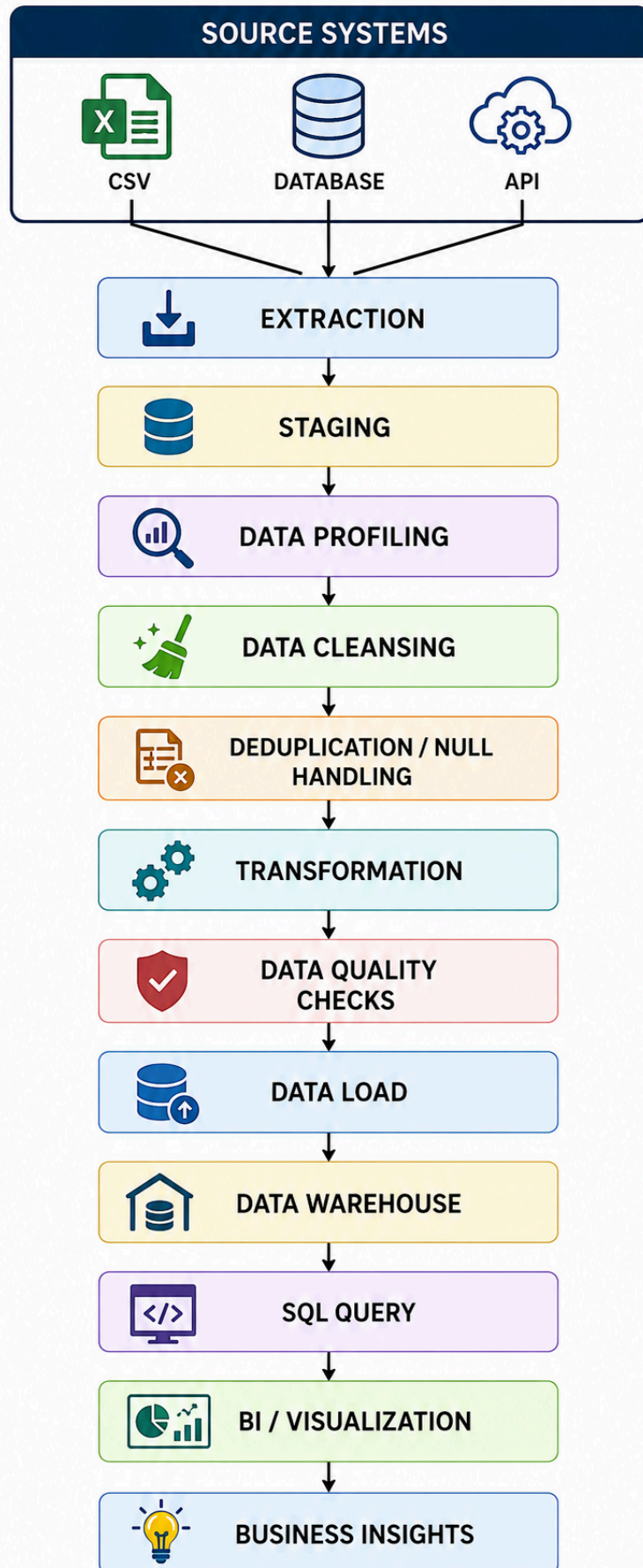
- Revenue by month

- Revenue by region
- Sales by product category
- Top 10 customers
- Top 10 products

Interactive Features

- Date filter
 - Region filter
 - Product filter
 - Customer segment filter
 - Drill-down
 - Cross-filtering
-

End-to-End Capstone Flow



Learning Outcome

By the end of this module, the trainee should be able to:

Extract → Transform → Validate → Load → Query → Visualize → Analyze

and understand how a real-world data engineering pipeline connects **source systems**, **ETL/ELT**, **data quality**, **cloud compute**, **data warehouses**, **SQL** and **BI**.

Transformation Logic

- **Apply consistent formats** — Ensure data follows a common structure and format across all sources.
- **Normalize units & measures** — Convert values into standardized units, such as **kg → grams** or **USD → INR**.
- **Standardize date formats** — Convert different date representations into one standard format, such as **YYYY-MM-DD**.
- **Apply coding schemes** — Map inconsistent codes and values to standardized business codes.
- **Enhance readability** — Clean column names, text values, labels, and descriptions for easier interpretation.
- **Align with reporting rules** — Apply business definitions, calculations, filters, and aggregation rules required for reporting.

Simple flow:

Raw Data → Format Standardization → Unit Normalization → Date Standardization → Coding → Cleansing → Business Rules → Reporting-Ready Data